ELECTRICAL & COMPUTER ENGINEERING
TEXAS A&M UNIVERSITY

A Wideband Millimeter-Wave CMOS Receiver Using Reconfigurable Low-Noise Amplifier With a 3-Winding Transformer Load

Mahdi Rajaei

Outline

Introduction

ELECTRICAL & COMPUTER ENGINEERING
TEXAS A&M UNIVERSITY

- 5th Generation wireless communication:
  - High Speed, Reliability, and efficiency
  - Further increase the capacity of wireless communication
  - Supporting more devices
  - Supporting new cases: Smart Cities, Autonomous Vehicles, Entertainment, Agriculture

Introduction

- 5G wireless communication systems (Frequency)
- Sub-6G
  - 410 – 7100 MHz, including multiple sub-bands.
- mm-wave bands
  - N257: 26.5 – 29.5 GHz
  - N258: 24.25 – 27.5 GHz
  - N259: 39.5 – 43.5 GHz
  - N260: 37 – 40 GHz
  - N261: 27.5 – 28.35 GHz

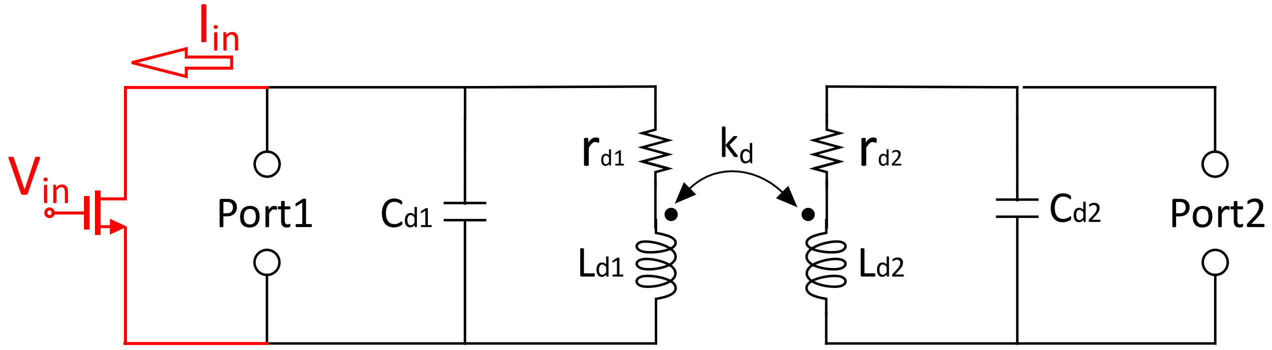
Main Contributions

Wideband and Reconfigurable Frequency mode of Operation

Interference Signal Cancelling After LNA

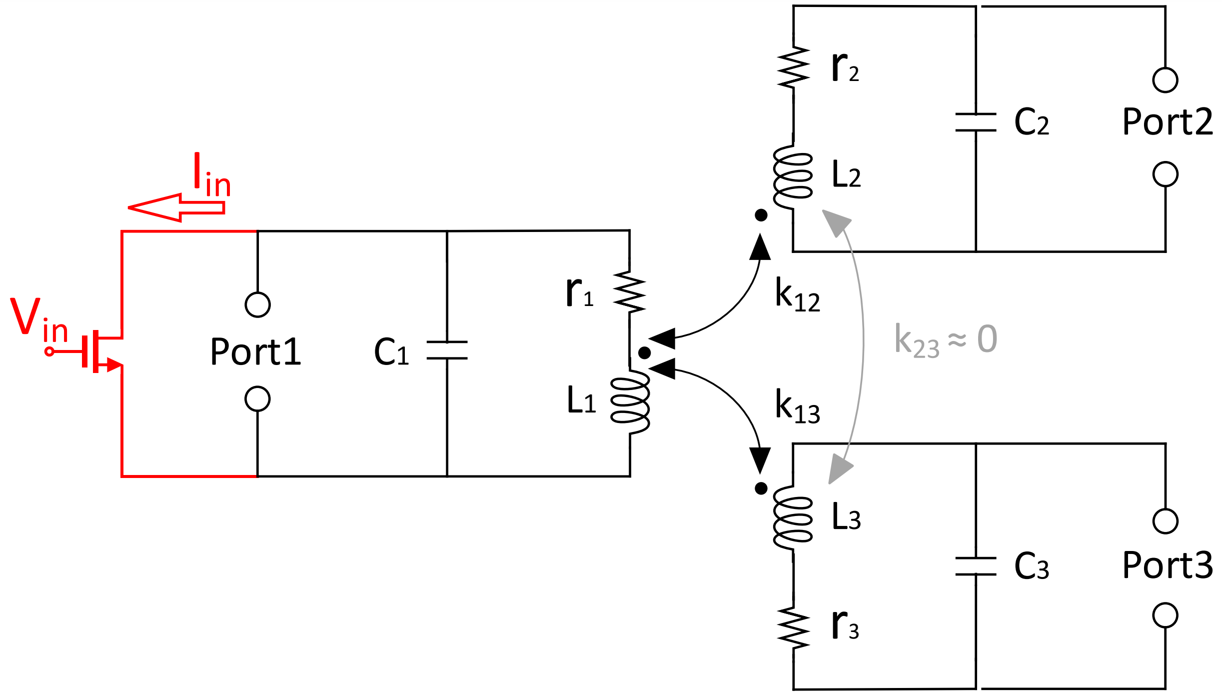
On-Chip Image Rejection Using 3-Stage Filter

Double-Tuned and 3-Winding Transformer Loads



$$Gain_1(j\omega) = g_m Z_{11}(j\omega)$$

$$Gain_2(j\omega) = g_m Z_{21}(j\omega)$$



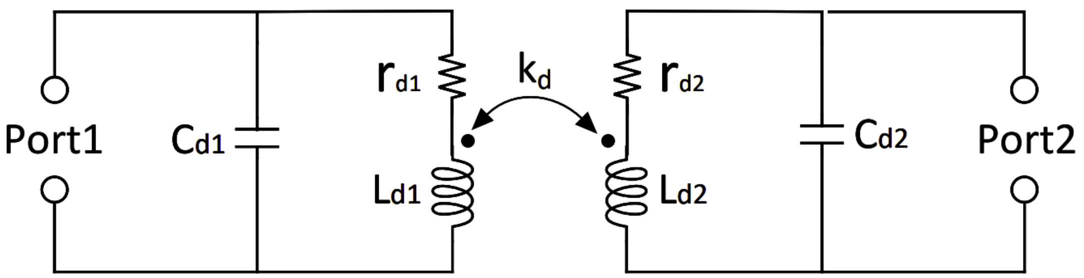
$$Gain_1(j\omega) = g_m Z_{11}(j\omega)$$

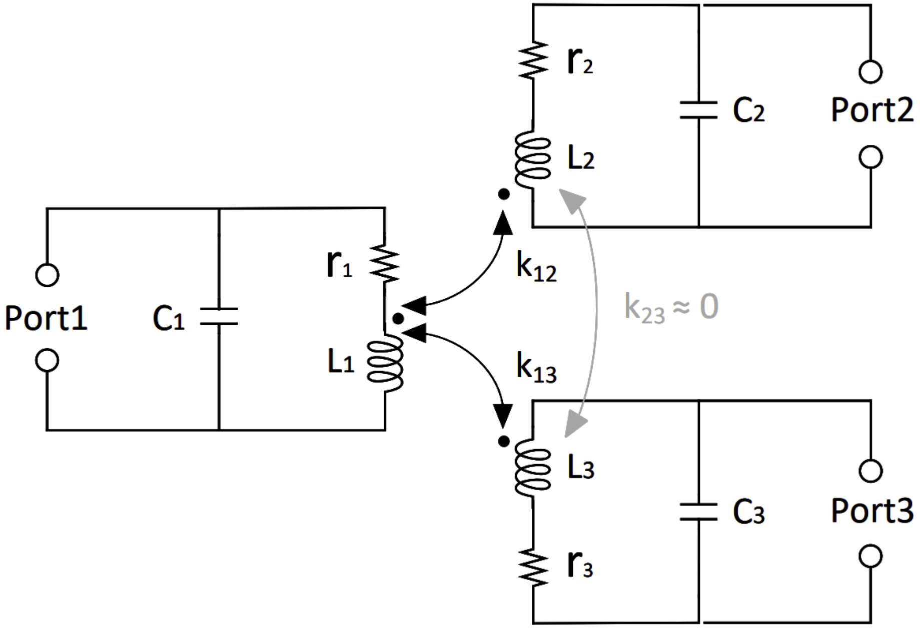
$$Gain_2(j\omega) = g_m Z_{21}(j\omega)$$

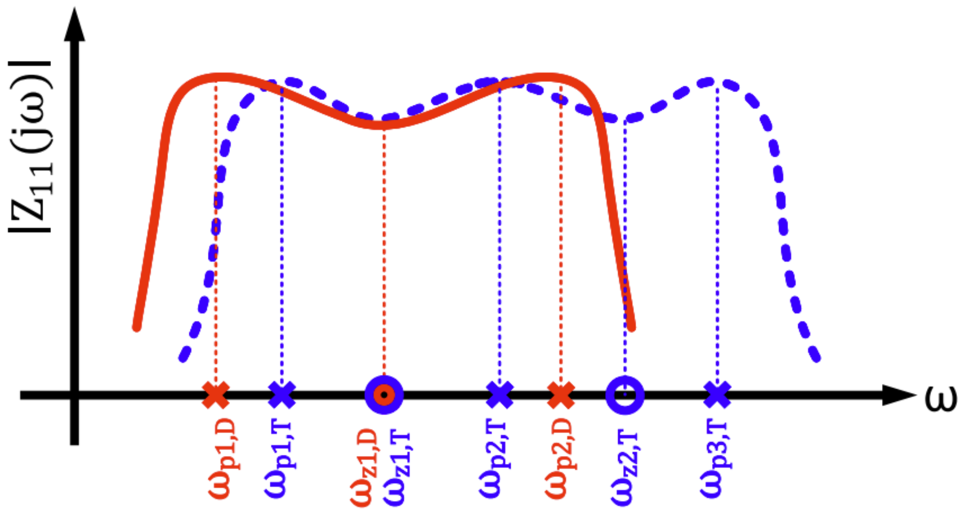
$$Gain_3(j\omega) = g_m Z_{31}(j\omega)$$

3-Winding Transformer Load

ELECTRICAL & COMPUTER ENGINEERING
TEXAS A&M UNIVERSITY

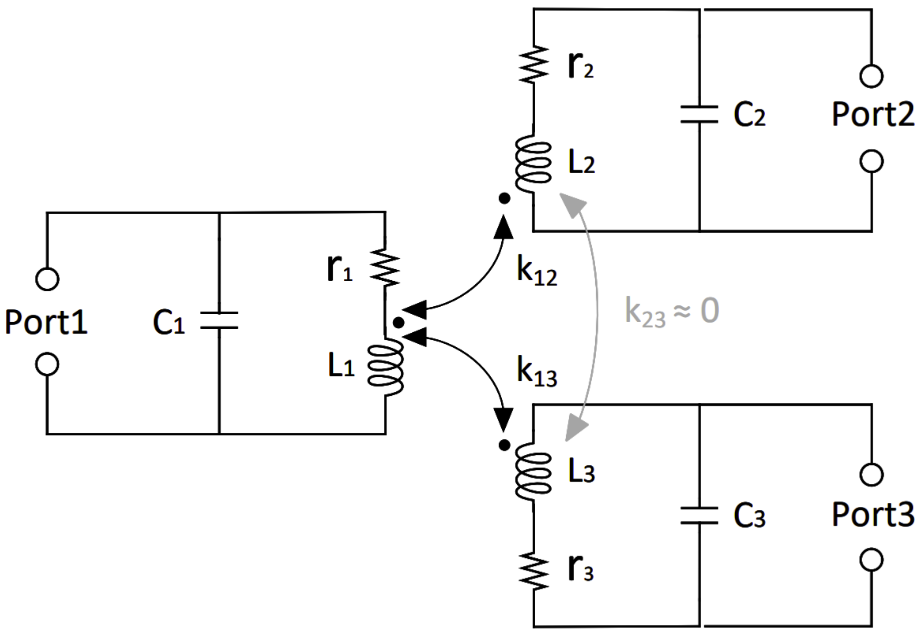






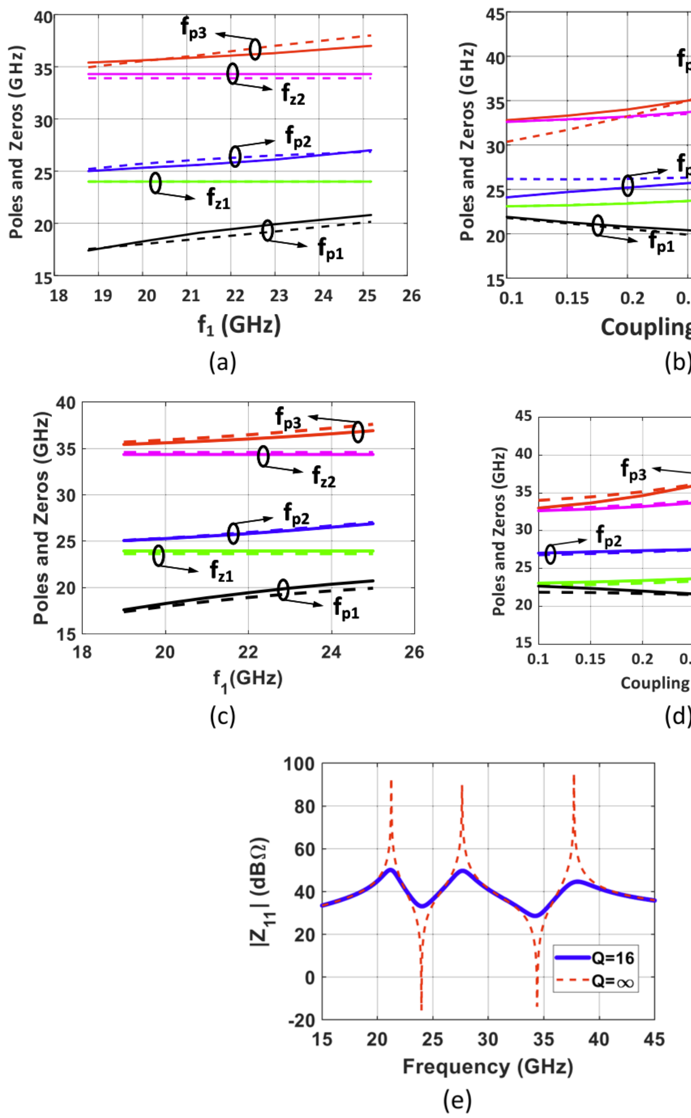
Double-Tuned Transformer	Triple-Coupled Transformer
$\omega_{p1,D} = \dfrac{\omega_1}{\sqrt{1+k}}$	$\omega_{p1,T} = \omega_1\sqrt{1-k}$
$\omega_{p2,D} = \dfrac{\omega_1}{\sqrt{1-k}}$	$\omega_{p2,T} = \omega_2 \dfrac{\left(\dfrac{\omega_1}{\omega_{p1,T}}\right)\left(\dfrac{\omega_3}{\omega_{p3,T}}\right)}{\sqrt{1-2k^2}}$
$\omega_{z1,D} = \omega_1$	$\omega_{p3,T} = \omega_3\sqrt{1+\sqrt{2}k}$
	$\omega_{z1,T} = \omega_1$ $\omega_{z2,T} = \omega_2$

3-Winding Transformer Load



Double-Tuned Transformer	Triple-Coupled Transformer
$\omega_{p1,D} = \dfrac{\omega_1}{\sqrt{1+k}}$	$\omega_{p1,T} = \omega_1\sqrt{1-k}$
$\omega_{p2,D} = \dfrac{\omega_1}{\sqrt{1-k}}$	$\omega_{p2,T} = \omega_2 \dfrac{\left(\dfrac{\omega_1}{\omega_{p1,T}}\right)\left(\dfrac{\omega_3}{\omega_{p3,T}}\right)}{\sqrt{1-2k^2}}$
$\omega_{z1,D} = \omega_1$	$\omega_{p3,T} = \omega_3\sqrt{1+\sqrt{2}k}$
	$\omega_{z1,T} = \omega_1$
	$\omega_{z2,T} = \omega_2$

solid-line: ADS model dash-line: Equation

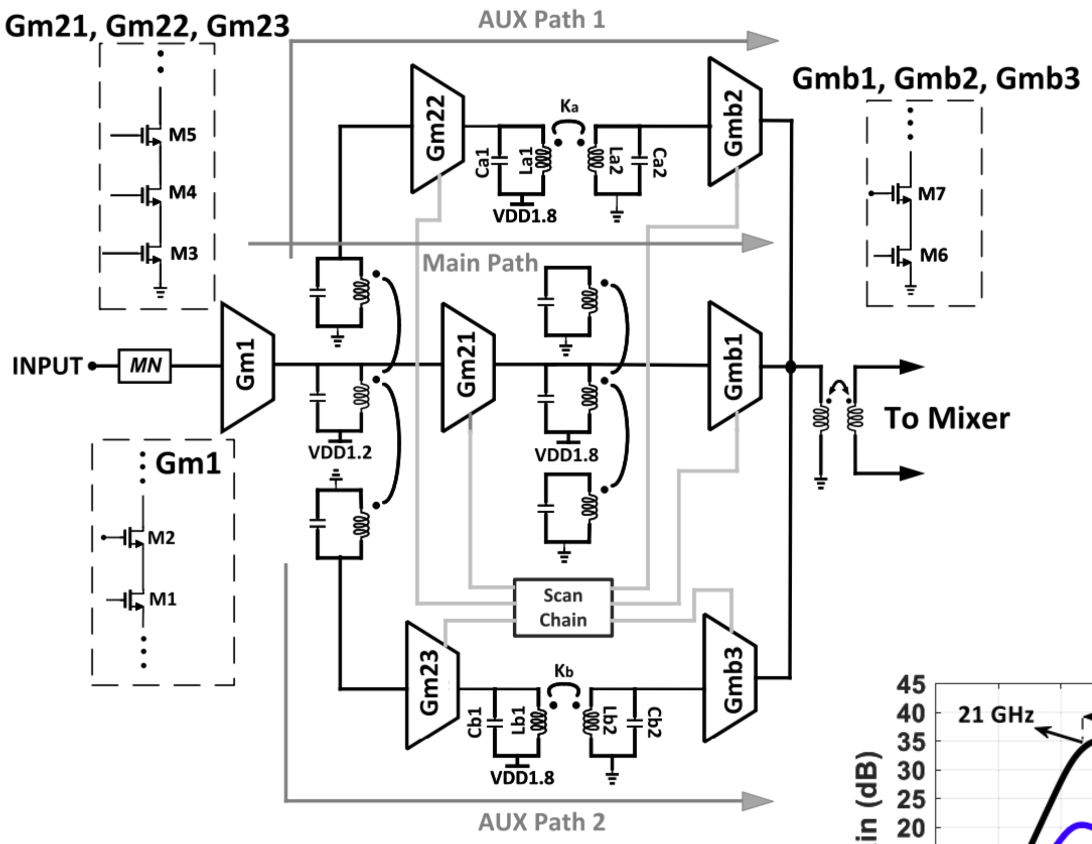


$Q = \infty$

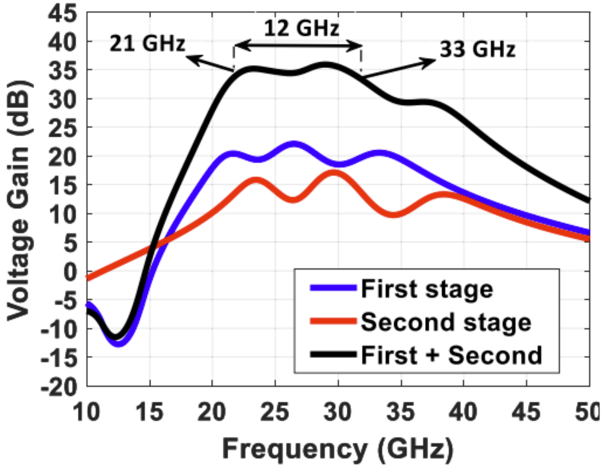
$Q = 16$

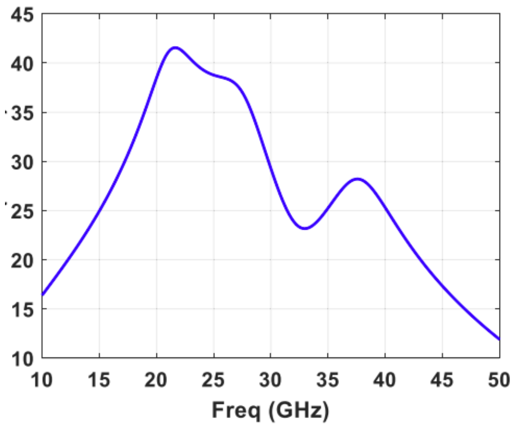
Wideband/Reconfigurable LNA

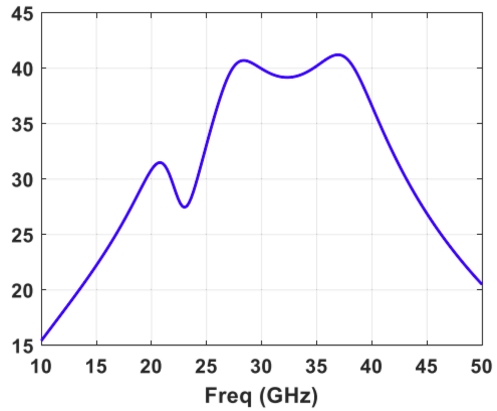
ELECTRICAL & COMPUTER ENGINEERING
TEXAS A&M UNIVERSITY



Mode	Main Path	AUX1 Path	AUX2 Path	Active Output Port	Frequency Range (GHz)	Peak Gain (dB)	Functionality
Wideband Mode	✅ ON	❌ OFF	❌ OFF	Port 1 (Z11)	21.5 – 32.5	45.5 @ 22.5	Covers full bandwidth with flat gain
Low-Frequency Mode	❌ OFF	✅ ON	❌ OFF	Port 2 (Z21)	21.5 – 26.5	45.45 @ 22	Enhances low-frequency operation, cancels high-frequency interference
High-Frequency Mode	❌ OFF	❌ OFF	✅ ON	Port 3 (Z31)	27 – 32	46.58 @ 29	Enhances high-frequency operation, cancels low-frequency interference

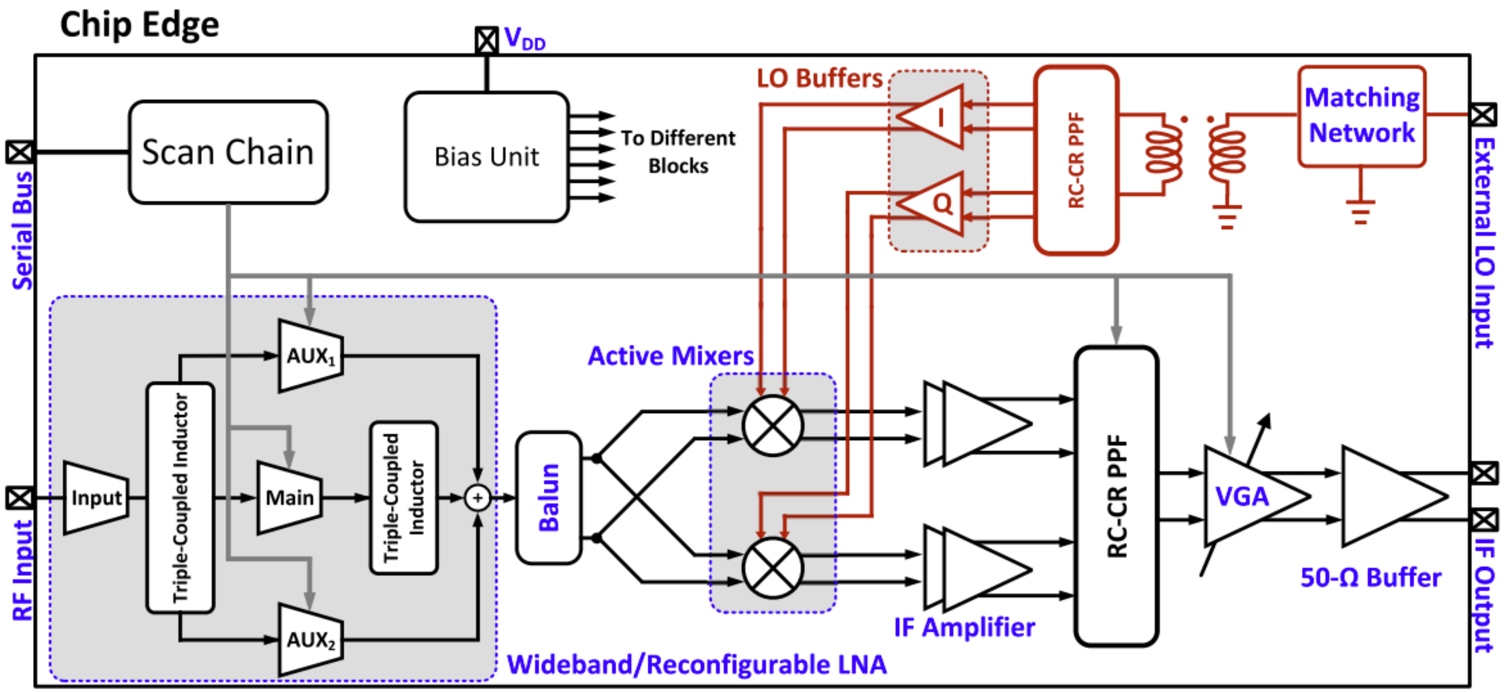


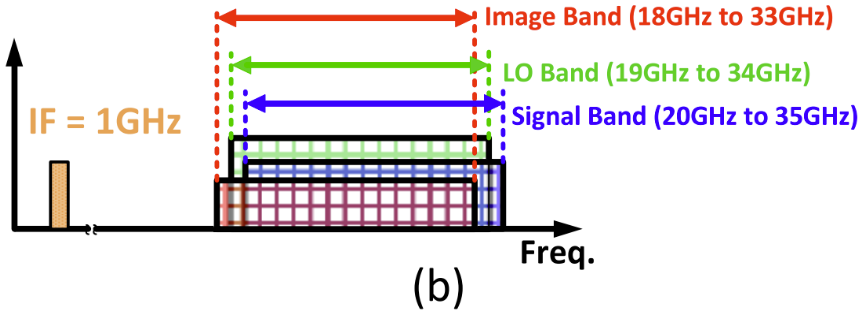


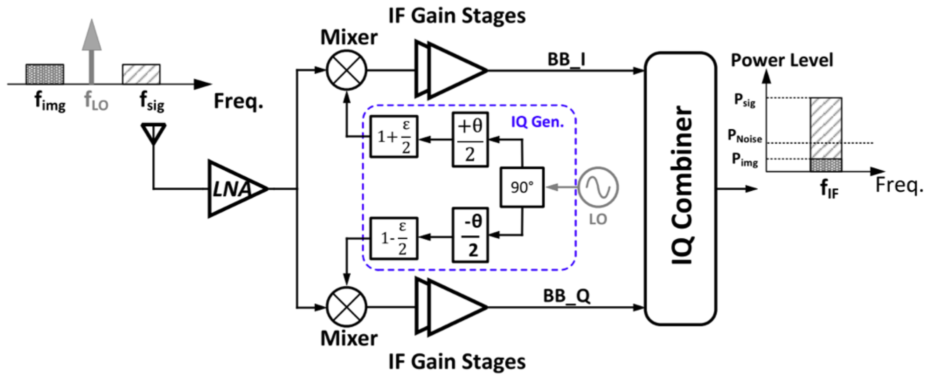


Receiver Architecture

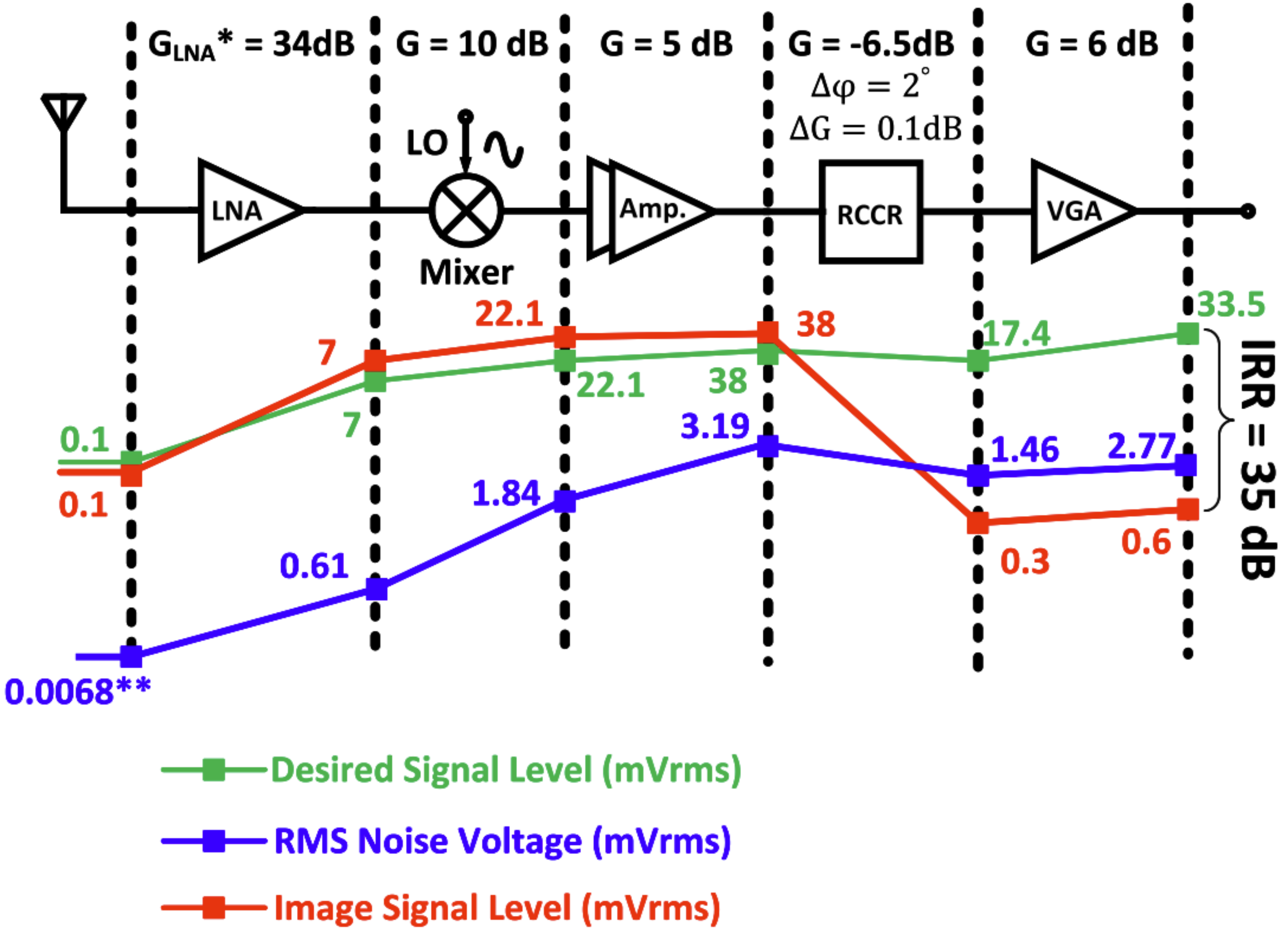
ELECTRICAL & COMPUTER ENGINEERING
TEXAS A&M UNIVERSITY



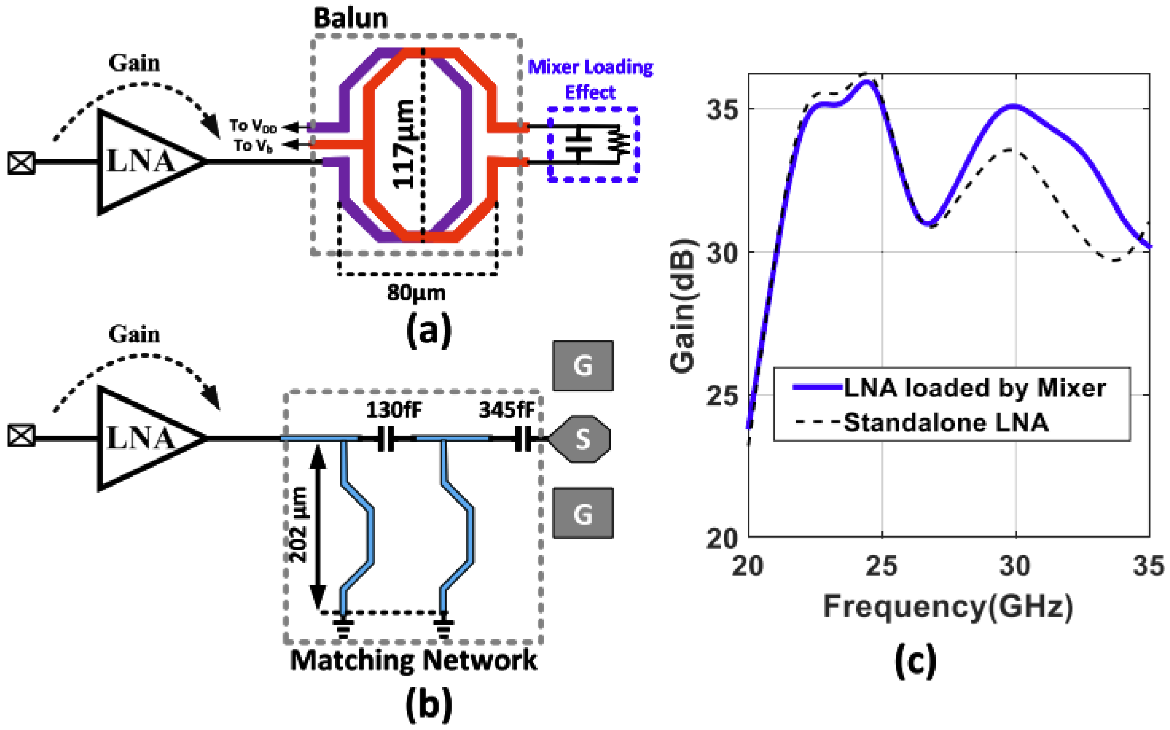


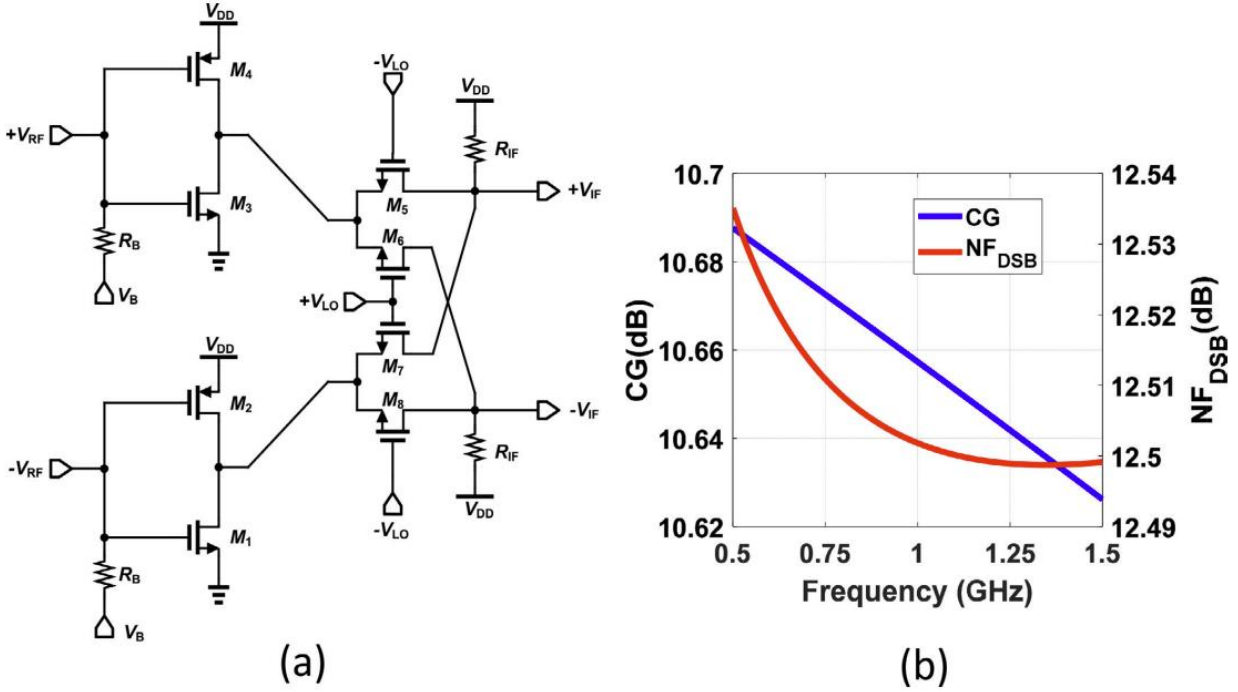


Receiver Architecture

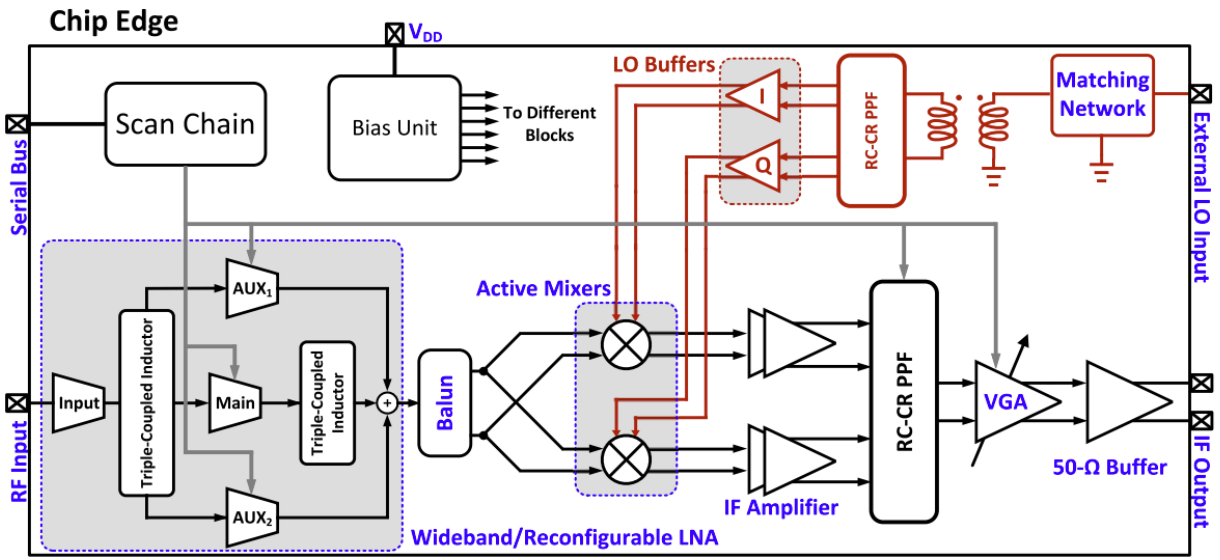


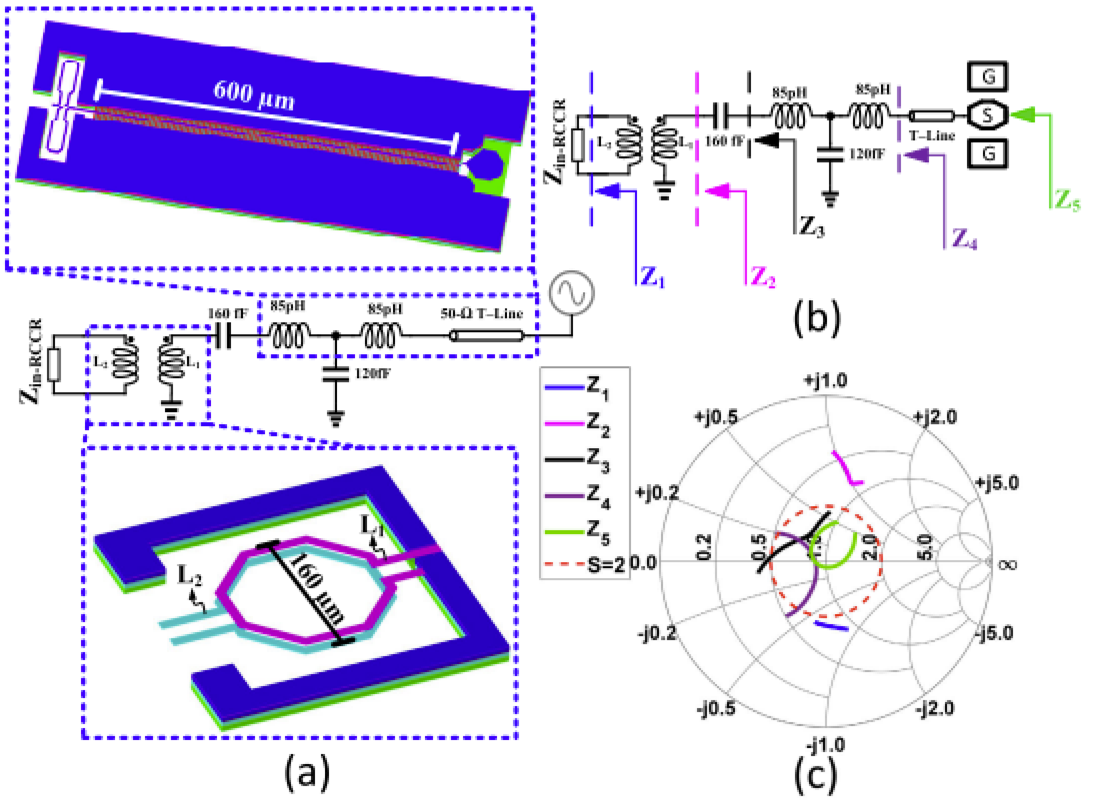
RF Gm-Boosted Double-Balanced Mixers



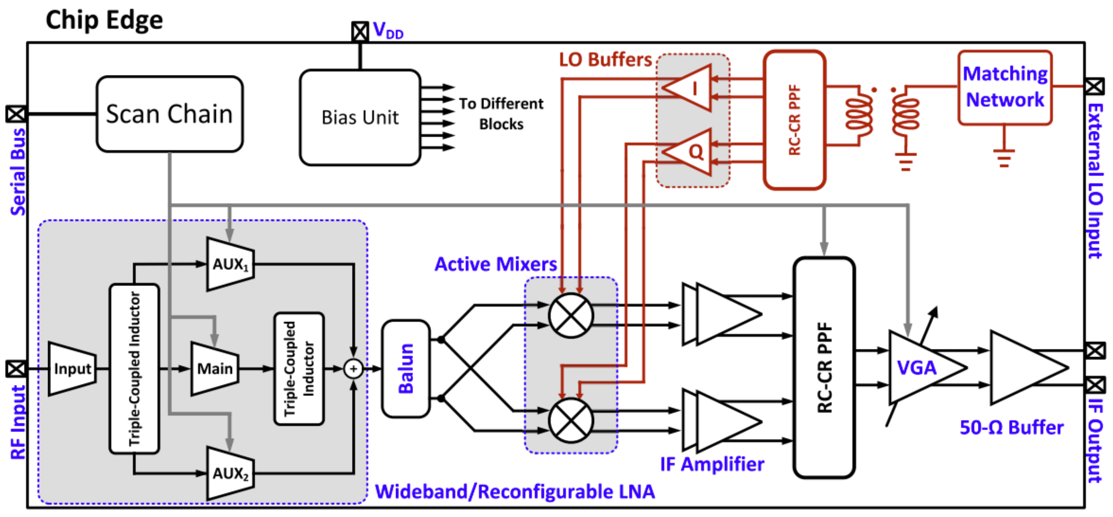


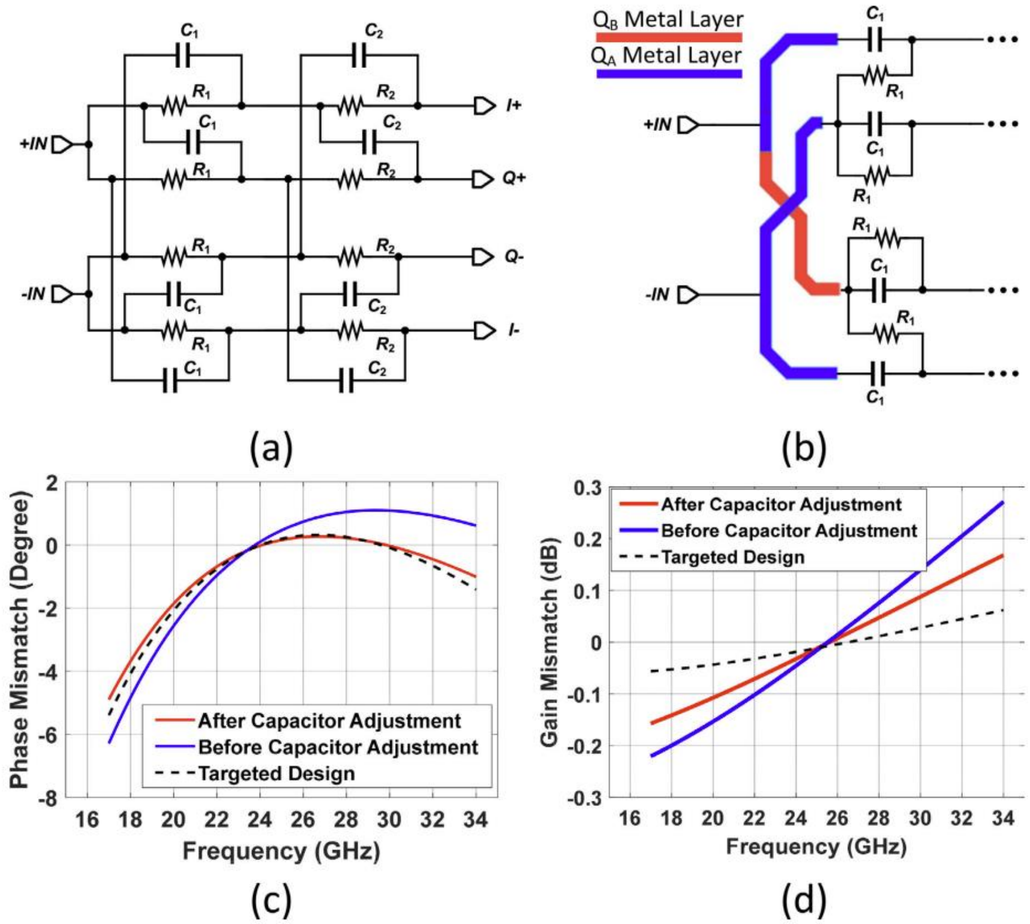
LO Path


Chip Edge
VDD
Scan Chain
Bias Unit
To Different Blocks
LO Buffers
I
Q
RC-CR PPF
Matching Network
External LO Input
Serial Bus
Active Mixers
AUX1
Input
Triple-Coupled Inductor
Main
Triple-Coupled Inductor
AUX2
Balun
RC-CR PPF
VGA
IF Amplifier
50-Ω Buffer
IF Output
RF Input
Wideband/Reconfigurable LNA

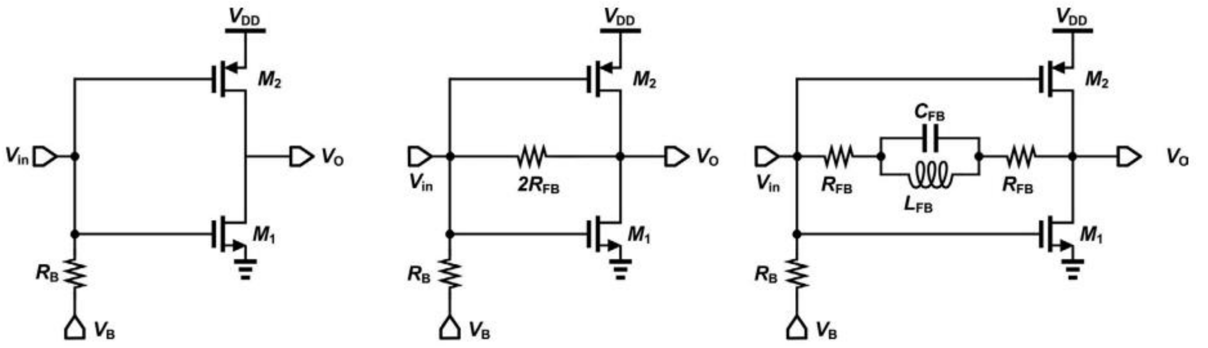

600 µm
Zin-RCCR
160 fF
85pH
85pH
50-Ω T-Line
120fF
L2
L1
160 µm
G
S
G
T-Line
Z1
Z2
Z3
Z4
Z5
S=2
+j1.0
+j0.5
+j2.0
+j0.2
+j5.0
-j0.2
-j5.0
-j0.5
-j2.0
-j1.0
(a)
(b)
(c)

LO Path

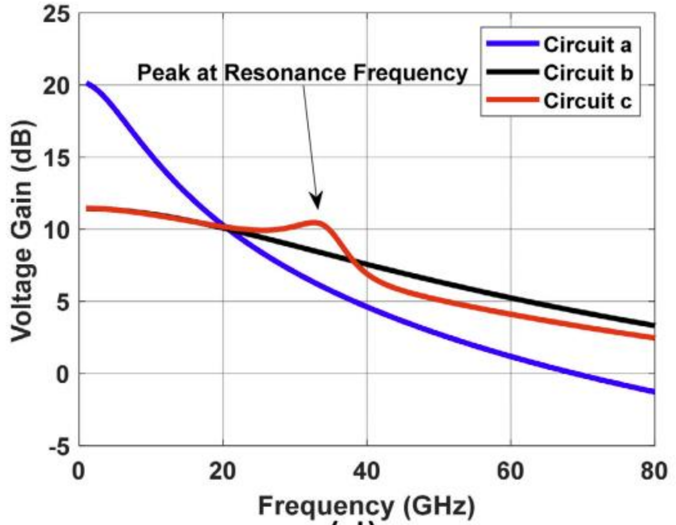

Chip Edge
V_{DD}
Scan Chain
Bias Unit
To Different Blocks
LO Buffers
I
Q
RC-CR PPF
Matching Network
External LO Input
Serial Bus
RF Input
Input
Triple-Coupled Inductor
AUX_1
Main
AUX_2
Triple-Coupled Inductor
Balun
Active Mixers
IF Amplifier
RC-CR PPF
VGA
50-Ω Buffer
IF Output
Wideband/Reconfigurable LNA


+IN
-IN
C_1
C_2
R_1
R_2
I+
Q+
Q-
I-
(a)
Q_B Metal Layer
Q_A Metal Layer
(b)
Phase Mismatch (Degree)
Frequency (GHz)
After Capacitor Adjustment
Before Capacitor Adjustment
Targeted Design
(c)
Gain Mismatch (dB)
Frequency (GHz)
After Capacitor Adjustment
Before Capacitor Adjustment
Targeted Design
(d)

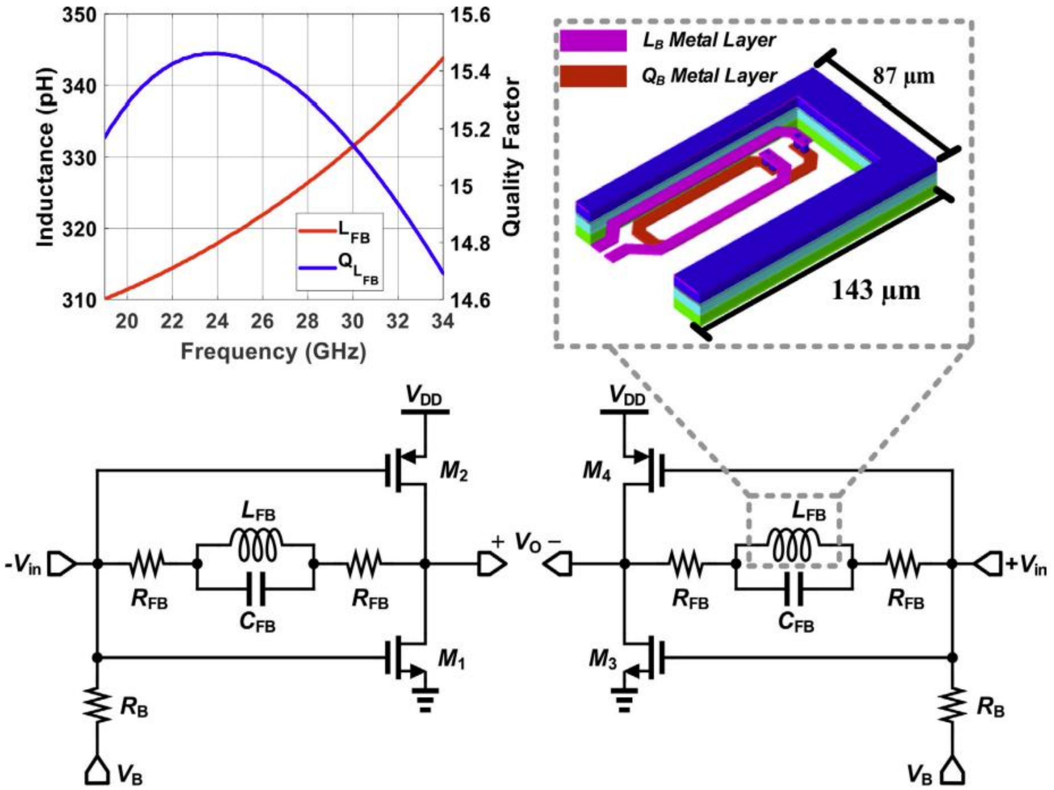
LO Buffers



(a) (b) (c)

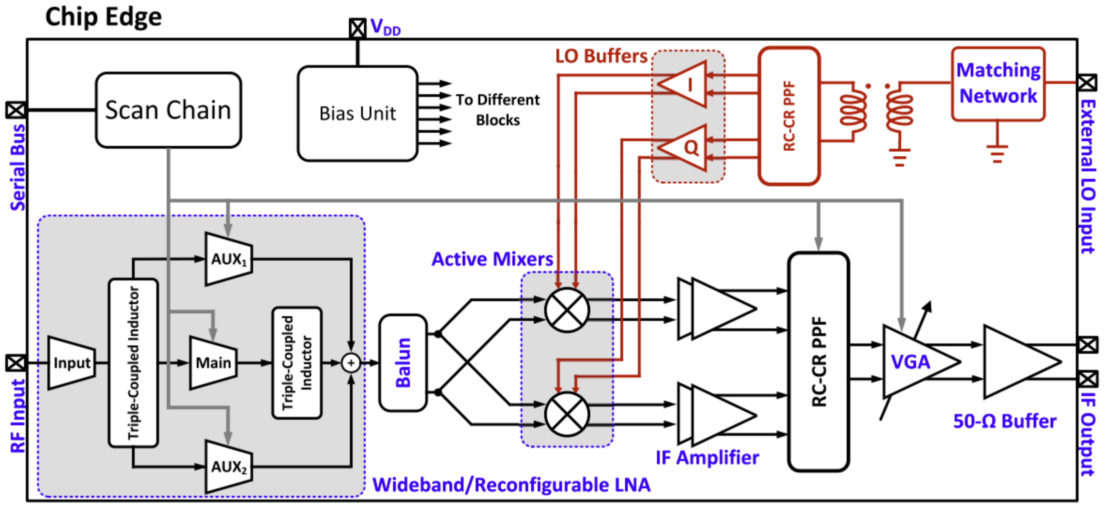


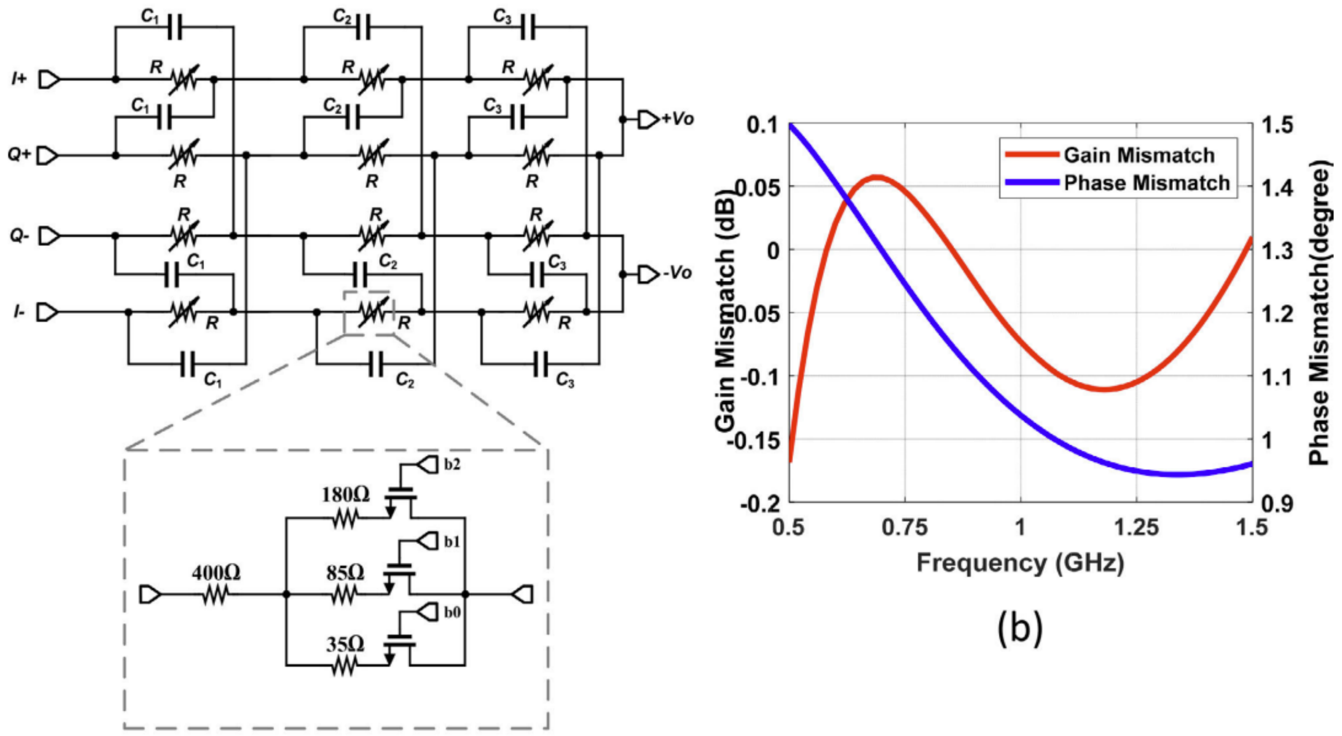
(d)



IF Circuits

ELECTRICAL & COMPUTER ENGINEERING
TEXAS A&M UNIVERSITY



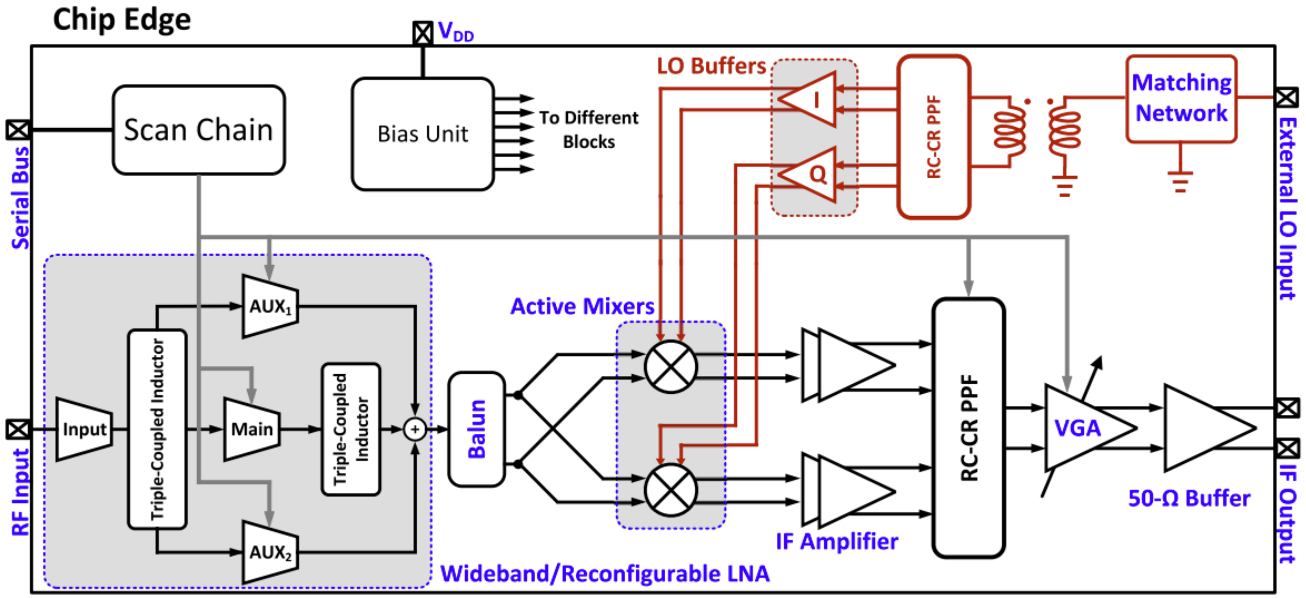


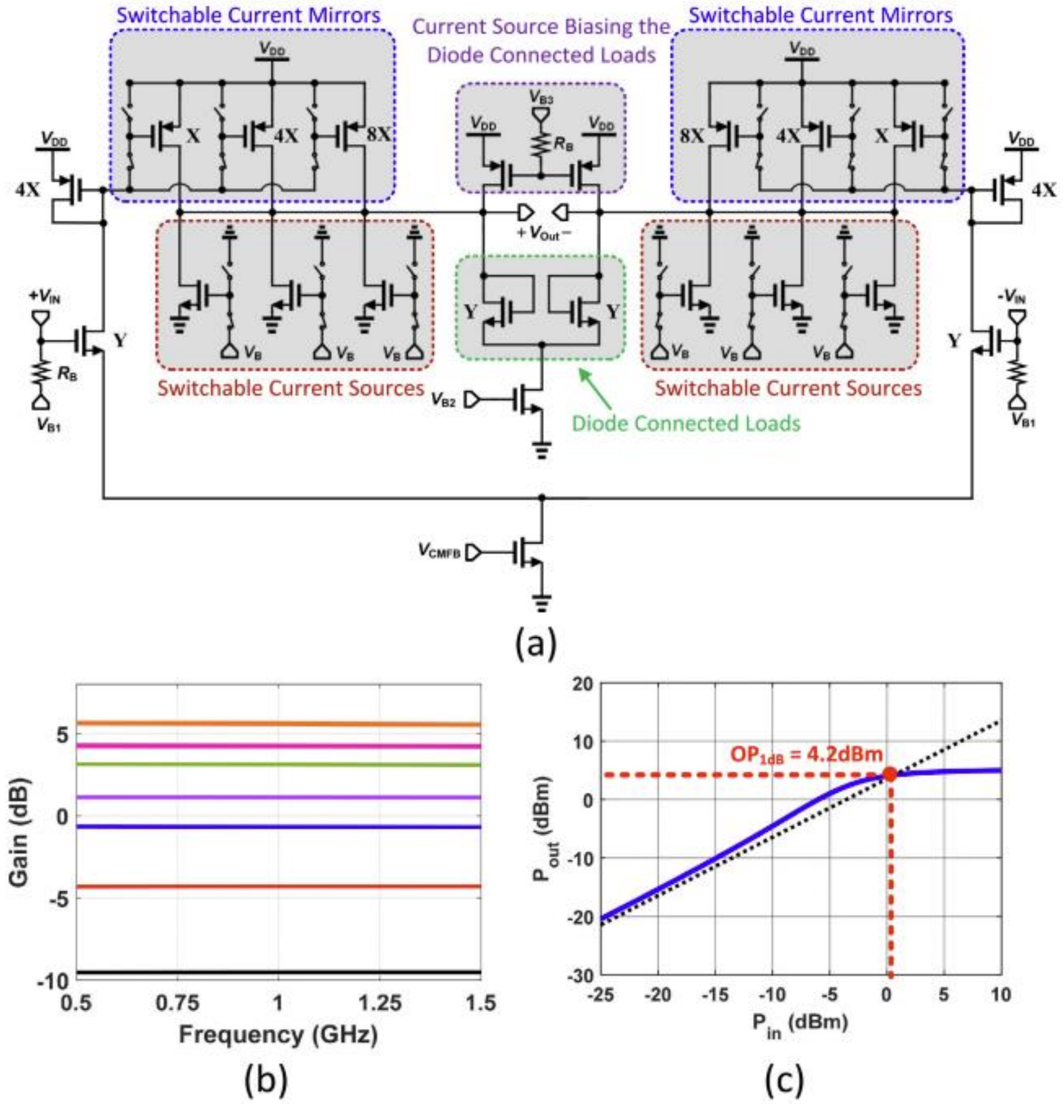
(a)

(b)

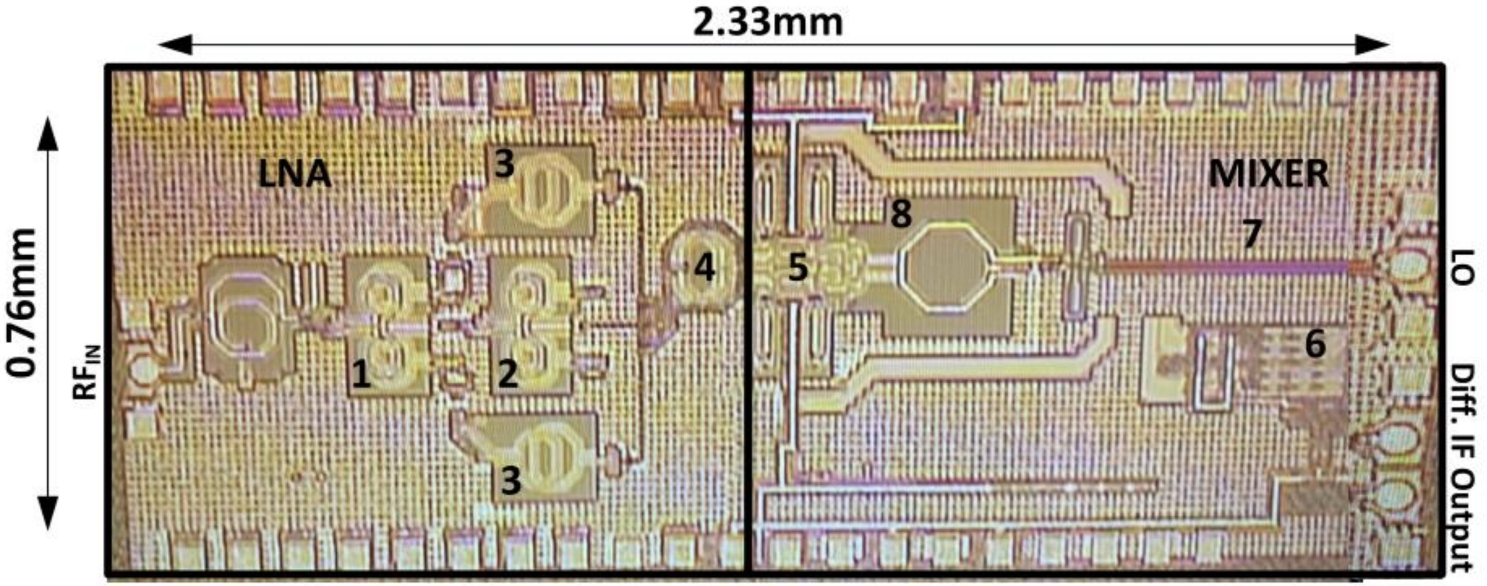
IF Circuits

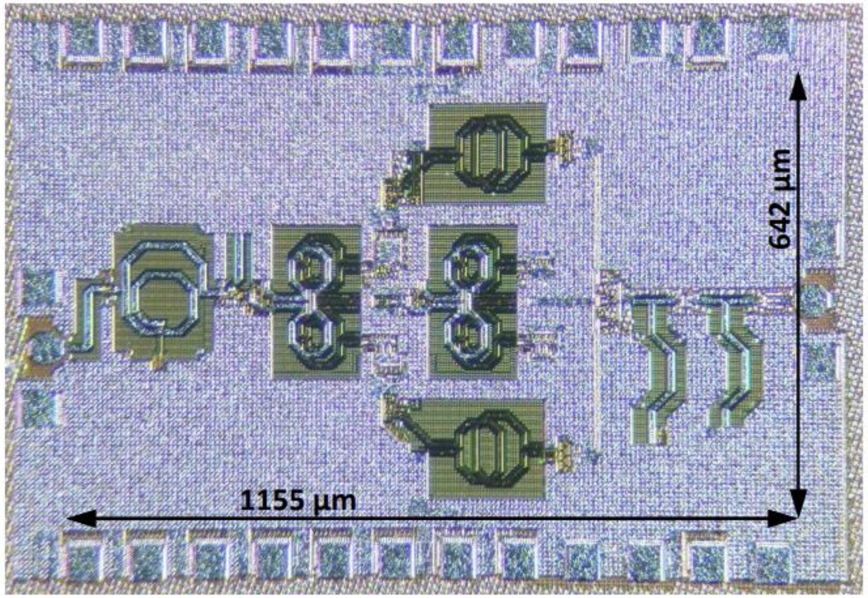
ELECTRICAL & COMPUTER ENGINEERING
TEXAS A&M UNIVERSITY





Fabrication and Measurements





The Chips have been Fabricated Using CMOS 22nm CMOS FDSOI

Fabrication and Measurements

ELECTRICAL & COMPUTER ENGINEERING
TEXAS A&M UNIVERSITY

PERFORMANCE COMPARISON WITH THE STATE-OF-THE-ART 3-WINDING MM-WAVE LNAS

	[14] RFIC'23	[11] JSSC'24	[12] TMTT'24	[13] TCAS'23
Process	22nm FDSOI	40nm CMOS	65nm CMOS	130nm CMOS SOI
Frequency (GHz)	21.6-34.2	51.6-73.7	32-46	6-12
Freq. Operation Mode	WB[b]/RC[c]	WB[b]	WB[b]	WB[b]
% FBW	45	35.3	35.9	66.7
Minimum NF (dB)	2.3	3.78	2.2	0.8
Peak Gain (dB)	32.4	22.4	21.5	23.3
Min. OP_{1dB} (dBm)	-6	7.2[a]	-	-1.7[a]
Power (mW)	35	34	22	66
Area (mm^2)	0.74	0.12	0.16	1.41

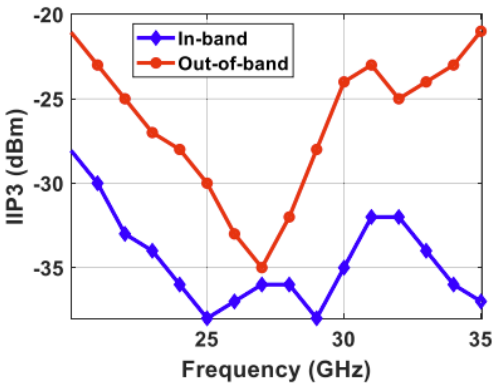
[a] Approximated as ($Gain_{dB}$ - 1) + IP_{1dB} when measured result not available. [b] Wideband. [c] Re-configurable

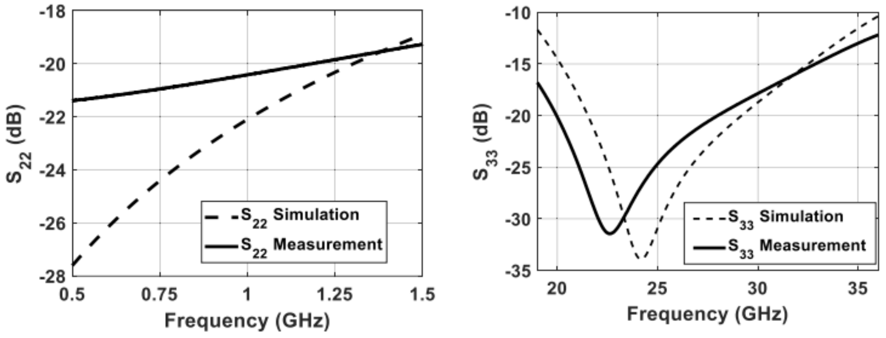
PERFORMANCE COMPARISON WITH THE STATE-OF-THE-ART MM-WAVE WIDEBAND RECEIVERS

Specs \ References	This work		[6] MTT'20	[3][c] MTT'20	[7] EuMIC'18	[4] JSSC'20	[2] MTT'2021	[9] JSSC'19
Technology	22nm CMOS FDSOI		65nm CMOS	22nm CMOS FDSOI	130nm ST SiGe BiCMOS	45nm CMOS SOI	22nm CMOS FDSOI	45nm CMOS SOI
on-chip image-rejection	YES		NO	YES	NO	YES	NO	YES
Freq operation mode	WB/RC*		WB*	WB*	WB*	WB*	WB*	WB/RC*
Freq (GHz)	21.5 – 32.5 [WB]		26.5 – 32.5	20 – 44	24 – 30	24.5 – 43.5	19.5 – 42	27 – 29.75/ 28.75 – 35
	21.5-26.5[RC]	27-32[RC]						
% FBW	40.7		20.34	75	22.2	55.9	73.2	19.6
CG (dB)	45.5		29.5[a]	28.5[a]	39.5	35.2	25.3	33/26.5
IRR (dB)	> 30		-	> 75	> 18	> 32	-	> 30
NF (dB)	5.5 – 2.9		5.3 – 6.6	3.3 – 5	3.6 – 4.8	3.2 – 6.1	2.7 – 4.2	5.7/8.5
OP1dB (dBm)	-2.8 – 2		0.5[b]	-1 – 3[b]	15	9 – 27	-1.7 – 1.3[b]	2/2.5
Power (mW)	94.8		33	70	1270	60	102	52.5
Area (mm^2)	1.78 (pad excluded)		0.65	1.8	4.05	0.77	0.9	0.46

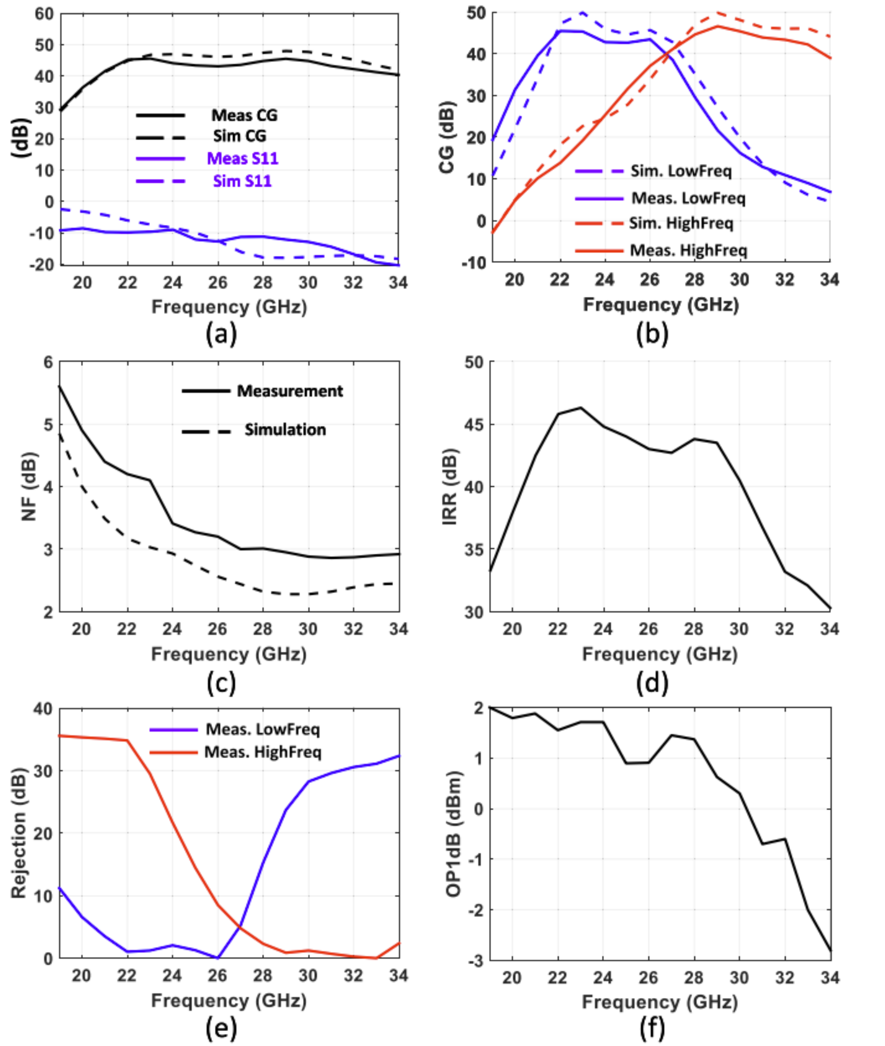
[a] Output IF amplifier and IQ mixer gain are excluded. [b] OP1dB = (CG_{dB} – 1) + iP1dB
[c] IQ mixer and output IF amplifier are off-chip. * WB: Wideband RC:Reconfigurable

Fabrication and Measurements





(a) (b)



POWER CONSUMPTION DETAILS

Component	Power consumption [mW]	Power Supply Voltage [V]
LNA	35	1.2/1.8
Mixers	21.6	1.2
LO Buffers	15.84	1.2
VGA	3.9	1.2
IF Amplifier	15.2	1.2
Bias Circuitry	3.3	1.2
Receiver	94.8	1.2/1.8

ELECTRICAL & COMPUTER ENGINEERING
TEXAS A&M UNIVERSITY

Thanks!